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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-061

1. Topic (12 words max)

Dengue Outbreak Prediction Using Machine Learning and Weather Data

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-061
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Dengue is a major public health problem in the tropics as well as subtropics, even in Sri Lanka, as biannual breakouts affect the healthcare structure of the respective countries to a great extent [1], [2]. Rainfall, temperature, and humidity are important climate variables for the breeding of mosquitoes and the spread of dengue virus; hence, dengue is a highly climate-sensitive disease [3]. Although historical dengue surveillance data and meteorological records are available through government authorities, dengue control efforts in Sri Lanka remain largely reactive, with preventive measures typically initiated only after a substantial increase in reported cases [2].

The current method of dengue surveillance is based on descriptive statistics and retrospective analysis of trends [5]. Although these two methods of analysis can offer useful information regarding past trends of dengue outbreak patterns, neither of these methods can provide predictive analysis or early warning systems. This directly means that there are chances of missing opportunities regarding proactive prevention and strategic resource allocation in the healthcare sector. The transmission of dengue cases has been found to show complicated non-linear relationships between variables of climatic factors, which cannot be analyzed properly by statistics [3].

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Recent research has demonstrated the potential of machine learning techniques to model such non-linear and temporal relationships in disease prediction tasks [4]. Algorithms such as Random Forests, gradient boosting models, and neural networks have shown improved predictive performance compared to traditional methods. However, many existing dengue prediction studies are limited to national-level analyses or offline experimental models that are not tailored to localized settings such as districts or administrative regions [4], [5]. In addition, the lack of transparent and interpretable prediction outputs reduces trust and limits the practical adoption of these systems by public health decision-makers.

Another critical gap in existing research is the limited integration of advanced data engineering, explainable machine learning, and language-based intelligence within a unified predictive framework. Without proper data preprocessing, spatio-temporal alignment, and explainability, even accurate machine learning models are not easily implementable. Thus, there exists an evident need to develop a machine learning-based dengue outbreak prediction system that uses historical dengue patient data as well as weather data to offer local, explainable, and actionable early alerts to enable data-informed dengue control efforts in Sri Lanka.

References

[1] World Health Organization, *Dengue and Severe Dengue*. Geneva, Switzerland: WHO, 2023.

[2] National Dengue Control Unit, Sri Lanka, *Weekly Dengue Surveillance Reports*, Ministry of Health, 2022–2024.

[3] A. Rocklöv and L. Johansson, "The impact of climate change on dengue transmission," *PLoS Neglected Tropical Diseases*, vol. 13, no. 1, 2019.

[4] C. Xu *et al.*, "Machine learning models for dengue outbreak prediction using climate variables," *International Journal of Environmental Research and Public Health*, vol. 17, no. 12, 2020.

[5] A. G. L. Perera and M. D. Fernando, "Spatiotemporal analysis of dengue incidence in Sri Lanka," *BMC Public Health*, vol. 21, no. 1, 2021.

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6. Brief description of Existing Research and Systems (300 words max)

Most of the current work done on dengue outbreak prediction has focused on analyzing the correlation of historical dengue incidence with meteorological variables such as rainfall, temperature, and humidity. Many works have used traditional statistical and time-series methods, such as linear regression models, ARIMA models, and seasonal trend analysis to find temporal patterns and seasonality in dengue transmission [5]. These methods are useful for retrospective analysis, but quite a few researchers comment about the limitations of these techniques to portray the rich nonlinear and lagged dependencies typical in mosquito breeding cycles and in environmental conditions [3].

To overcome these shortcomings, researchers have gradually shifted their focus to applying machine learning models in predicting cases of dengue infection. Various machine learning algorithms, including Random Forest, SVM models, and Gradient Boosting models, as well as Artificial Neural Network models, have been employed to capture the non-linear association between climatic variables and the spread of dengue infection [4]. However, most existing studies focus on national or large regional datasets, which limits their ability to capture localized spatial variations in dengue risk, particularly at district or sub-district levels [5].

Furthermore, many of the proposed machine learning models are evaluated in offline experimental settings and are not integrated into operational decision-support systems. As a result, their practical applicability for real-time public health planning remains limited [4]. Another commonly identified shortcoming is the lack of transparency and interpretability in prediction outputs. Several studies acknowledge that machine learning models often function as black boxes, providing risk predictions without explaining the contributing factors, which reduces trust and hinders adoption by public health authorities [3].

Additionally, existing systems predominantly focus on numerical prediction outcomes and often neglect the integration of advanced data engineering practices, explainable analytics, and language-based intelligence. Lack of interaction and interpretation interfaces makes the system even more restricted for non-technical people, such as those contributing to dengue prevention and control. In summary, even though proof of the feasibility of machine learning-based dengue outbreak prediction is available from previous work, there are still considerable gaps regarding localized prediction and integration.

References

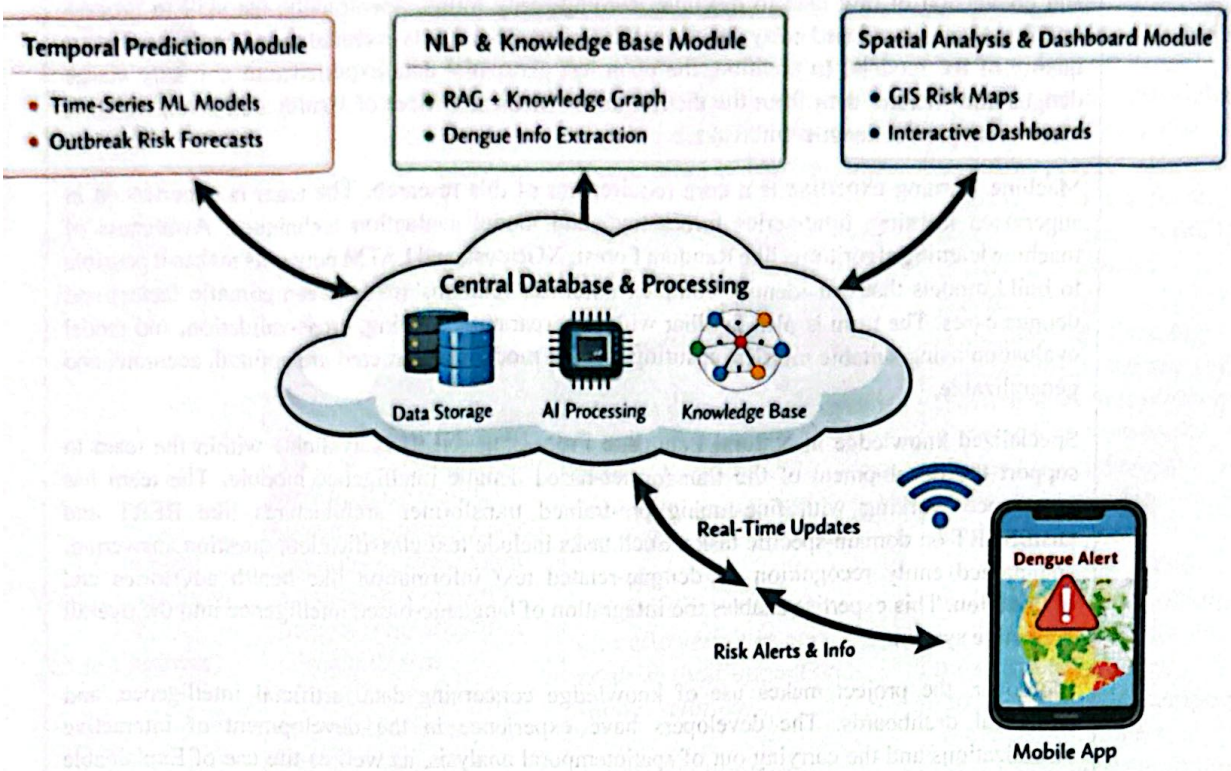
[1] World Health Organization, *Dengue and Severe Dengue*. Geneva, Switzerland: WHO, 2023.
[2] National Dengue Control Unit, Sri Lanka, *Weekly Dengue Surveillance Reports*, Ministry of Health, 2022–2024.
[3] A. Rocklöv and L. Johansson, "The impact of climate change on dengue transmission," *PLoS Neglected Tropical Diseases*, vol. 13, no. 1, 2019.
[4] C. Xu et al., "Machine learning models for dengue outbreak prediction using climate variables," *International Journal of Environmental Research and Public Health*, vol. 17, no. 12, 2020.
[5] A. G. L. Perera and M. D. Fernando, "Spatiotemporal analysis of dengue incidence in Sri Lanka," *BMC Public Health*, vol. 21, no. 1, 2021.

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7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

- The designed solution for these issues is a dengue outbreak prediction system developed based on machine learning approaches that would be effectively useful for early warning and proactive decision-making for dengue prevention in Sri Lanka in particular, focusing on the Colombo District. The selection of the Colombo District is made because of its highest incidence of dengue cases, maximum populations, as well as suitable climatic conditions for breeding. The main aim of the system would be to predict risks of possible dengue outbreaks by processing past dengue cases along with some weather-related parameters.
- The process relies on two broad categories of inputs. The data collected from the first category comprises historical data from cases of dengue, which has been collected from the National Dengue Control Unit. The data collected comprises inputs such as date, geographic locations, and the number of cases recorded. The second type of data collected comprises historical data on weather conditions, which has been collected from the Department of Meteorology. The data comprises inputs such as rain, temperature, humidity, and speed of the winds. All these help determine the dynamics between the mosquitoes and the spread of the disease.
- The solution follows a structured, multi-stage machine learning pipeline. Initially, data collection is performed by aggregating dengue and weather datasets from the respective institutions. This is followed by a comprehensive data preprocessing phase, which includes data cleaning, handling missing values, normalization, temporal alignment, and feature engineering. Time-lagged features are generated to capture delayed effects between climatic conditions and dengue outbreaks, reflecting real-world mosquito life cycles.
- Once the data is ready, machine learning algorithms are used to train the models to identify the complex and nonlinear relationships between the weather and the dengue cases. Various machine learning algorithms, such as Random Forest, XGBoost, and LSTM networks, are used to ensure that the model has good predictive capabilities. Evaluations are carried out by using various metrics to determine which machine learning model would work best for prediction.
- The models have been capable of producing predictions regarding dengue breakouts in the form of risk levels, classified as Low, Medium, or High, on a weekly basis. The system is also capable of producing data that helps define the risk differentiated according to space, specifically for the Colombo District.
  With the emphasis on local prediction, incorporating various weather Factors, and leveraging smart machine learning methods, the proposed system is developed to fill the identified research gaps. Additionally, a conceptual system diagram highlighting the entire data flow process, right from input data sources, preprocessing, training, prediction, to output, will also be presented in the proposed system.

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8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

A multidisciplinary skill set including data engineering, machine learning, natural language processing, and advanced analytics is needed to ensure the effective implementation of the proposed machine learning-based dengue outbreak prediction system. Fortunately, the research team is collectively endowed with the necessary skills and knowledge to design and implement the various components of the proposed system.

Data engineering skill and data preprocessing skill are essential to process the diverse dengue and meteorological data. The team has experience with data collection, preprocessing, normalization, and conversion of raw data to machine learning-ready forms. Specifically, the skill to process missing, inconsistent, and noisy data from the real environment is essential to ensure the predictive quality of the models. In addition, the team has geospatial data experience to correctly merge dengue and weather data from the district to the Medical Officer of Health area level, which is essential to predict dengue outbreaks.

Machine learning expertise is a core requirement of this research. The team is experienced in supervised learning, time-series forecasting, and model evaluation techniques. Awareness of machine learning algorithms like Random Forest, XGBoost, and LSTM networks makes it possible to build models that can identify complex nonlinear relationships between climatic factors and dengue cases. The team is also familiar with hyperparameter tuning, cross-validation, and model evaluation using suitable metrics, ensuring that the models constructed are optimal, accurate, and generalizable.

Specialized knowledge in Natural Language Processing (NLP) is available within the team to support the development of the transformer-based dengue intelligence module. The team has experience working with fine-tuning pre-trained transformer architectures like BERT and DistilBERT on domain-specific tasks. Such tasks include text classification, question answering, and named entity recognition on dengue-related text information like health advisories and information. This expertise enables the integration of language-based intelligence into the overall predictive system.

Moreover, the project makes use of knowledge concerning data, artificial intelligence, and analytical dashboards. The developers have experience in the development of interactive visualizations and the carrying out of spatiotemporal analysis, as well as the use of Explainable AI, such as feature importance analysis and SHAP explanations. This ensures that the results of machine learning predictions from the system can be interpreted effectively for decision-making by health organizations.

Regarding data availability, the project utilizes reliable sources. Past data on dengue cases will be collected from the **National Dengue Control Unit (NDU) of Sri Lanka**, and there will be no doubt about the reliability and authenticity of data on epidemiology. Meteorological data such as rainfall, temperature, humidity, and speed of the wind will be collected from the **Department of Meteorology in Sri Lanka**, with the use of satellite data on the climatic conditions when necessary. This availability of data confirms the ability and sustainability of this proposed project as feasible research. The diverse expertise and availability of data confirm that this proposed project is possible and within the capability of the research team of this student.

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9. Objectives and Novelty

Main Objective
To design and develop a comprehensive machine learning–based dengue outbreak prediction system for the Colombo District by utilizing historical dengue case data and meteorological variables, with the objective of generating accurate, localized, and interpretable early warnings that support proactive dengue prevention and informed public health decision-making.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22243980 Wickramarachchi T.T. (Data Engineer)	To design, analyze, and optimize a reliable, spatially and temporally consistent dataset that accurately represents dengue outbreaks and climate-driven transmission dynamics for machine learning based prediction.	• Collect historical dengue case data from the National Dengue Control Unit and weather data from the Department of Meteorology. • Perform exploratory **data analysis (EDA)** to identify trends, seasonality, data quality issues, and noise. • Clean datasets by handling missing values, inconsistencies, and outliers using multiple preprocessing strategies. • Normalize and scale climatic variables and analyze their impact on model stability. • Engineer biologically meaningful **time-lag features (1–4 weeks)** based on mosquito life cycle dynamics. • Apply feature selection techniques (correlation analysis, mutual information, and model-based feature importance) to identify the most influential climatic and temporal features.	This component is novel because it treats data engineering as a standalone research problem rather than a routine preprocessing step. Instead of using a fixed pipeline, it experimentally evaluates multiple preprocessing, normalization, time-lag **feature engineering, and feature selection strategies** to determine how each choice affects dengue outbreak prediction performance. The component introduces biologically informed time-lag features based on mosquito life cycles, reducing arbitrary temporal assumptions. It also conducts spatial resolution sensitivity analysis at district and MOH levels to study geographic effects on prediction reliability. By emphasizing data quality, feature relevance, biological alignment, and spatial consistency, this component significantly improves model robustness and interpretability, an aspect rarely examined in depth in

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		• Perform **geospatial data linking** at district and MOH area levels and analyze the impact of spatial resolution on prediction accuracy. • Produce and document **multiple ML-ready dataset** variants for experimental evaluation.	existing dengue prediction research.
IT22165848 Hettiarachchi A.T. (Data Visualization & Research Integration Developer)	To design an advanced analytics and explainable dashboard that transforms machine learning predictions into interpretable and actionable insights.	• **Design interactive dashboards** to visualize dengue risk levels across districts and time periods. • Develop **temporal trend and climate–dengue correlation visualizations.** • Integrate machine learning prediction outputs into the dashboard. • Apply **Explainable AI techniques** such as **feature importance and SHAP to explain** individual predictions. • Enable filtering, comparison, and drill-down analysis by district and time. • Perform **spatial analysis** to identify **geographic dengue risk patterns** across districts and MOH areas. • Develop **GIS-based spatial visualizations** such as risk **heatmaps and choropleth maps** to highlight high-risk regions.	Most existing dengue prediction systems function as **black-box models** that provide risk predictions without meaningful explanations, limiting trust and real-world usability. This component introduces **Explainable AI (XAI) techniques** such as feature importance analysis and **SHAP explanations** to make machine learning predictions transparent and interpretable **for public health decision-makers.** In addition, it integrates interactive dashboards with **spatial analysis** to visually identify where dengue risk is geographically concentrated across **districts and MOH areas** and how it evolves over time. By combining explainability, interactivity, and spatial intelligence within a single analytical platform, this component transforms raw model outputs into **actionable, evidence-based insights,** significantly enhancing both research novelty and practical applicability an approach rarely

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			implemented in existing dengue surveillance systems.
IT22214034 Keenavinna K.A.I.U. (NLP Engineer)	To design and implement a **Knowledge-Augmented Natural Language Processing framework** that integrates transformer-based intelligence with a dynamic Dengue Knowledge Base to provide evidence-based risk communication, verifiable insights, and automated surveillance reporting.	• Establish a **multi-modal dengue knowledge base** by collecting and curating high-quality textual data such as national health guidelines, clinical research publications, public advisories, and localized news articles. • Store semantic representations of dengue-related content in a **Vector Database** (e.g., ChromaDB or Pinecone) to enable efficient similarity-based information retrieval. • Design and populate a **Dengue Knowledge Graph** using a Graph Database (e.g., Neo4j) to model relationships between symptoms, mosquito breeding patterns, environmental triggers, and geographic risk zones. • Fine-tune transformer-based models (BERT / DistilBERT / GPT-based models) on dengue-specific content to improve domain understanding. • Implement **Retrieval-Augmented Generation (RAG)** to generate grounded and verifiable responses by combining transformer models with the vector knowledge store, reducing medical misinformation and hallucinations.	Unlike standard rule-based systems or purely generative chatbots, this component introduces a **hybrid intelligence layer** by integrating transformer-based language models with structured knowledge engineering through vector and graph databases. By anchoring NLP outputs to a dengue-specific knowledge base and employing Retrieval-Augmented Generation, the system delivers **medically accurate, explainable, and verifiable intelligence**. This approach bridges the gap between numerical outbreak predictions and human-interpretable knowledge, creating a centralized digital dengue intelligence framework tailored to the Sri Lankan public health context—an area rarely explored in existing dengue surveillance systems.

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		• Develop automated NLP pipelines to extract emerging dengue-related trends from news and public sources, updating a **surveillance database** to support early-warning signals. • Perform **cross-database synthesis** by linking NLP-derived insights with historical dengue case data and climatic information to generate structured health bulletins and situational reports.	
IT22281982 Adithya A.A.H (ML Engineer)	To develop, evaluate, and optimize machine learning models capable of **predicting dengue outbreak risk levels and outbreak probabilities in advance** using historical dengue surveillance and climatic data.	• Formulate dengue outbreak prediction as both a **time-based classification problem** (Low / Medium / High risk) and a **probabilistic forecasting task** to estimate outbreak likelihood. • Perform exploratory analysis on temporal dengue patterns to identify seasonality and outbreak cycles relevant to model design. • Train and compare multiple machine learning and deep learning models, including **Random Forest, XGBoost, and LSTM**, to capture non-linear and long-term temporal dependencies between weather variables and dengue cases. • Apply **time-series–aware data splitting strategies** (walk-forward validation and rolling windows) to prevent future data leakage and	This component is novel because it **systematically isolates and studies the temporal outbreak prediction problem**, focusing on *when* dengue risk will increase rather than attempting to jointly model spatial and temporal dynamics. Unlike many existing studies that rely on a single algorithm or broad national-level datasets, this work conducts a structured comparison between classical machine learning and deep learning time-series models using **localized Sri Lankan data.** The use of time-series–aware validation, probabilistic risk forecasting, and multi-horizon evaluation produces a robust, region-specific early warning mechanism. By emphasizing realistic forecasting behavior and comparative model analysis, this component provides practical and scientifically

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		ensure realistic forecasting performance. • Optimize model performance using **hyperparameter tuning techniques** such as grid search and random search. • Evaluate and compare models using appropriate classification and regression metrics including accuracy, recall, F1-score, ROC-AUC, and RMSE. • Analyze model stability across different forecasting horizons (1–4 weeks) and select the most reliable model or ensemble for final outbreak prediction.	grounded insights that go beyond conventional dengue prediction approaches.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Good Topic

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Shashika	Lokuliyana	26/1/12
Co-Supervisor	Ms.	Narmada	Gamage	26/1/12
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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